two computationally efficient algorithms for solving the normal equations. One algorithm, originally due to Levinson (1947) and modified by Durbin (1959), is called the Levinson–Durbin algorithm. This algorithm is suitable for serial processing and has a computation complexity of $O(p^2)$. The second algorithm, due to Schür (1917), also computes the reflection coefficients in $O(p^2)$ operations but with parallel processors, the computations can be performed in $O(p)$ time. Both algorithms exploit the Toeplitz symmetry property inherent in the autocorrelation matrix.

We begin by describing the Levinson–Durbin algorithm.

11.3.1 The Levinson–Durbin Algorithm

The Levinson-Durbin algorithm is a computationally efficient algorithm for solving the normal equations in (11.3.1) for the prediction coefficients. This algorithm exploits the special symmetry in the autocorrelation matrix

$$p = \begin{bmatrix} \gamma_{xx}(0) & \gamma_{xx}^*(1) & \cdots & \gamma_{xx}^*(p-1) \\ \gamma_{xx}(1) & \gamma_{xx}(0) & \cdots & \gamma_{xx}^*(p-2) \\ \vdots & \vdots & & \\ \gamma_{xx}(p-1) & \gamma_{xx}(p-2) & \cdots & \gamma_{xx}(0) \end{bmatrix} \tag{11.3.3}$$

Note that $\Gamma_p(i, j) = \Gamma_p(i - j)$, so that the autocorrelation matrix is a *Toeplitz matrix*. Since $\Gamma_p(i, j) = \Gamma_p^*(j, i)$, the matrix is also Hermitian.

The key to the Levinson–Durbin method of solution, that exploits the Toeplitz property of the matrix, is to proceed recursively, beginning with a predictor of order $m = 1$ (one coefficient) and then to increase the order recursively, using the lower-order solutions to obtain the solution to the next-higher order. Thus the solution to the first-order predictor obtained by solving (11.3.1) is

$$a_1(1) = -\frac{\gamma_{xx}(1)}{\gamma_{xx}(0)} \tag{11.3.4}$$

and the resulting MMSE is

$$\begin{aligned} E_1^f &= \gamma_{xx}(0) + a_1(1)\gamma_{xx}(-1) \\ &= \gamma_{xx}(0)[1 - |a_1(1)|^2] \end{aligned} \tag{11.3.5}$$

Recall that $a_1(1) = K_1$, the first reflection coefficient in the lattice filter.

The next step is to solve for the coefficients $\{a_2(1), a_2(2)\}$ of the second-order predictor and express the solution in terms of $a_1(1)$. The two equations obtained from (11.3.1) are

$$\begin{aligned} a_2(1)\gamma_{xx}(0) + a_2(2)\gamma_{xx}^*(1) &= -\gamma_{xx}(1) \\ a_2(1)\gamma_{xx}(1) + a_2(2)\gamma_{xx}(0) &= -\gamma_{xx}(2) \end{aligned} \tag{11.3.6}$$

By using the solution in (11.3.4) to eliminate $\gamma_{xx}(1)$, we obtain the solution

$$
\begin{aligned}
a_2(2) &= -\frac{\gamma_{xx}(2) + a_1(1)\gamma_{xx}(1)}{\gamma_{xx}(0)[1 - |a_1(1)|^2]} \\
&= -\frac{\gamma_{xx}(2) + a_1(1)\gamma_{xx}(1)}{E_1^f} \qquad (11.3.7) \\
a_2(1) &= a_1(1) + a_2(2)a_1^*(1)
\end{aligned}
$$

Thus we have obtained the coefficients of the second-order predictor. Again, we note that $a_2(2) = K_2$, the second reflection coefficient in the lattice filter.

Proceeding in this manner, we can express the coefficients of the mth-order predictor in terms of the coefficients of the $(m-1)$st-order predictor. Thus we can write the coefficient vector $\mathbf{a}_m$ as the sum of two vectors, namely,

$$
\mathbf{a}_m = \begin{bmatrix} a_m(1) \\ a_m(2) \\ \vdots \\ a_m(m) \end{bmatrix} = \begin{bmatrix} \mathbf{a}_{m-1} \\ \cdots \\ 0 \end{bmatrix} + \begin{bmatrix} \mathbf{d}_{m-1} \\ \cdots \\ K_m \end{bmatrix} \tag{11.3.8}
$$

where $\mathbf{a}_{m-1}$ is the predictor coefficient vector of the $(m-1)$st-order predictor and the vector $\mathbf{d}_{m-1}$ and the scalar K_m are to be determined. Let us also partition the $m \times m$ autocorrelation matrix Γ_{xx} as

$$
\Gamma_m = \begin{bmatrix} \Gamma_{m-1} & \boldsymbol{\gamma}_{m-1}^{b^*} \\ \boldsymbol{\gamma}_{m-1}^{bt} & \gamma_{xx}(0) \end{bmatrix} \tag{11.3.9}
$$

where $\boldsymbol{\gamma}_{m-1}^{bt} = [\gamma_{xx}(m-1) \quad \gamma_{xx}(m-2) \quad \cdots \quad \gamma_{xx}(1)] = (\boldsymbol{\gamma}_{m-1}^{b})^t$, the asterisk (*) denotes the complex conjugate, and $\boldsymbol{\gamma}_m^t$ denotes the transpose of $\boldsymbol{\gamma}_m$. The superscript b on $\boldsymbol{\gamma}_{m-1}$ denotes the vector $\boldsymbol{\gamma}_{m-1}^t = [\gamma_{xx}(1) \quad \gamma_{xx}(2) \quad \cdots \quad \gamma_{xx}(m-1)]$ with elements taken in reverse order.

The solution to the equation $\Gamma_m \mathbf{a}_m = -\boldsymbol{\gamma}_m$ can be expressed as

$$
\begin{bmatrix} \Gamma_{m-1} & \boldsymbol{\gamma}_{m-1}^{b^*} \\ \boldsymbol{\gamma}_{m-1}^{bt} & \gamma_{xx}(0) \end{bmatrix} \left\{ \begin{bmatrix} \mathbf{a}_{m-1} \\ 0 \end{bmatrix} + \begin{bmatrix} \mathbf{d}_{m-1} \\ K_m \end{bmatrix} \right\} = -\begin{bmatrix} \boldsymbol{\gamma}_{m-1} \\ \gamma_{xx}(m) \end{bmatrix} \tag{11.3.10}
$$

This is the key step in the Levinson–Durbin algorithm. From (11.3.10) we obtain two equations, namely,

$$
\Gamma_{m-1}\mathbf{a}_{m-1} + \Gamma_{m-1}\mathbf{d}_{m-1} + K_m \boldsymbol{\gamma}_{m-1}^{b*} = -\boldsymbol{\gamma}_{m-1} \tag{11.3.11}
$$

$$
\boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1} + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{d}_{m-1} + K_m \gamma_{xx}(0) = -\gamma_{xx}(m) \tag{11.3.12}
$$

Since $\Gamma_{m-1}\mathbf{a}_{m-1} = -\boldsymbol{\gamma}_{m-1}$, (11.3.11) yields the solution

$$
\mathbf{d}_{m-1} = -K_m \Gamma_{m-1}^{-1} \boldsymbol{\gamma}_{m-1}^{b^*} \tag{11.3.13}
$$

But $\boldsymbol{\gamma}_{m-1}^{b^*}$ is just $\boldsymbol{\gamma}_{m-1}$ with elements taken in reverse order and conjugated. Therefore, the solution in (11.3.13) is simply

$$\mathbf{d}_{m-1} = K_m \mathbf{a}_{m-1}^{b*} = K_m \begin{bmatrix} a_{m-1}^*(m-1) \\ a_{m-1}^*(m-2) \\ \cdot \\ a_{m-1}^*(1) \end{bmatrix} \tag{11.3.14}$$

The scalar equation (11.3.12) can now be used to solve for K_m. If we eliminate $\mathbf{d}_{m-1}$ in (11.3.12) by using (11.3.14), we obtain

$$K_m[\gamma_{xx}(0) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}^{b*}] + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1} = -\gamma_{xx}(m)$$

Hence

$$K_m = -\frac{\gamma_{xx}(m) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}}{\gamma_{xx}(0) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}^{b*}} \tag{11.3.15}$$

Therefore, by substituting the solutions in (11.3.14) and (11.3.15) into (11.3.8), we obtain the desired recursion for the predictor coefficients in the Levinson–Durbin algorithm as

$$a_m(m) = K_m = -\frac{\gamma_{xx}(m) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}}{\gamma_{xx}(0) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}^{b*}} = -\frac{\gamma_{xx}(m) + \boldsymbol{\gamma}_{m-1}^{bt}\mathbf{a}_{m-1}}{E_m^f} \tag{11.3.16}$$

$$\begin{aligned} a_m(k) &= a_{m-1}(k) + K_m a_{m-1}^*(m-k) \\ &= a_{m-1}(k) + a_m(m) a_{m-1}^*(m-k) \qquad \begin{array}{l} k = 1, 2, \ldots, m-1 \\ m = 1, 2, \ldots, p \end{array} \end{aligned} \tag{11.3.17}$$

The reader should note that the recursive relation in (11.3.17) is identical to the recursive relation in (11.2.21) for the predictor coefficients, obtained from the polynomials $A_m(z)$ and $B_m(z)$. Furthermore, K_m is the reflection coefficient in the mth stage of the lattice predictor. This development clearly illustrates that the Levinson–Durbin algorithm produces the reflection coefficients for the optimum lattice prediction filter as well as the coefficients of the optimum direct-form FIR predictor.

Finally, let us determine the expression for the MMSE. For the mth-order predictor, we have

$$\begin{aligned} E_m^f &= \gamma_{xx}(0) + \sum_{k=1}^{m} a_m(k)\gamma_{xx}(-k) \\ &= \gamma_{xx}(0) + \sum_{k=1}^{m} [a_{m-1}(k) + a_m(m) a_{m-1}^*(m-k)]\gamma_{xx}(-k) \\ &= E_{m-1}^f[1 - |a_m(m)|^2] = E_{m-1}^f(1 - |K_m|^2) \qquad m = 1, 2, \ldots, p \end{aligned} \tag{11.3.18}$$

where $E_0^f = \gamma_{xx}(0)$. Since the reflection coefficients satisfy the property that $|K_m| \le 1$, the MMSE for the sequence of predictors satisfies the condition

$$E_0^f \ge E_1^f \ge E_2^f \ge \cdots \ge E_p^f \tag{11.3.19}$$

This concludes the derivation of the Levinson–Durbin algorithm for solving the linear equations $\mathbf{\Gamma}_m \mathbf{a}_m = -\boldsymbol{\gamma}_m$, for $m = 0, 1, \ldots, p$. We observe that the linear equations have the special property that the vector on the right-hand side also appears as a vector in $\mathbf{\Gamma}_m$. In the more general case, where the vector on the right-hand side is some other vector, say $\mathbf{c}_m$, the set of linear equations can be solved recursively by introducing a second recursive equation to solve the more general linear equations $\mathbf{\Gamma}_m \mathbf{b}_m = \mathbf{c}_m$. The result is a *generalized Levinson–Durbin* algorithm (see Problem 11.12).

The Levinson–Durbin recursion given by (11.3.17) requires $O(m)$ multiplications and additions (operations) to go from stage m to stage $m+1$. Therefore, for p stages it takes on the order of $1+2+3+\cdots+p(p+1)/2$, or $O(p^2)$, operations to solve for the prediction filter coefficients, or the reflection coefficients, compared with $O(p^3)$ operations if we did not exploit the Toeplitz property of the correlation matrix.

If the Levinson–Durbin algorithm is implemented on a serial computer or signal processor, the required computation time is on the order of $O(p^2)$ time units. On the other hand, if the processing is performed on a parallel processing machine utilizing as many processors as necessary to exploit the full parallelism in the algorithm, the multiplications as well as the additions required to compute (11.3.17) can be carried out simultaneously. Therefore, this computation can be performed in $O(p)$ time units. However, the computation in (11.3.16) for the reflection coefficients takes additional time. Certainly, the inner products involving the vectors $\mathbf{a}_{m-1}$ and $\boldsymbol{\gamma}_{m-1}^b$ can be computed simultaneously by employing parallel processors. However, the addition of these products cannot be done simultaneously, but instead, require $O(\log p)$ time units. Hence, the computations in the Levinson–Durbin algorithm, when performed by p parallel processors, can be accomplished in $O(p \log p)$ time.

In the following section we describe another algorithm, due to Schür (1917), that avoids the computation of inner products, and therefore is more suitable for parallel computation of the reflection coefficients.

11.3.2 The Schür Algorithm

The Schür algorithm is intimately related to a recursive test for determining the positive definiteness of a correlation matrix. To be specific, let us consider the autocorrelation matrix $\mathbf{\Gamma}_{p+1}$ associated with the augmented normal equations given by (11.3.2). From the elements of this matrix we form the function

$$R_0(z) = \frac{\gamma_{xx}(1)z^{-1} + \gamma_{xx}(2)z^{-2} + \cdots + \gamma_{xx}(p)z^{-p}}{\gamma_{xx}(0) + \gamma_{xx}(1)z^{-1} + \cdots + \gamma_{xx}(p)z^{-p}} \tag{11.3.20}$$

and the sequence of functions $R_m(z)$ defined recursively as

$$R_m(z) = \frac{R_{m-1}(z) - R_{m-1}(\infty)}{z^{-1}[1 - R_{m-1}^*(\infty)R_{m-1}(z)]} \qquad m = 1, 2, \ldots \tag{11.3.21}$$